Critical Rydberg electrometry with a light–matter phase transition

what Ding *et al.* and Wang *et al.* do, with the transition of Zhang *et al.* in place of the interaction-driven one

Feasibility study and literature assessment

September 3, 2026

Abstract

Ding *et al.* [Nat. Phys. **18**, 1447 (2022)] and Wang *et al.* [arXiv:2502.19761] measure microwave electric fields by biasing a Rydberg ensemble at a critical point and reading the steep response. Their critical point is that of a *non-equilibrium* transition — the optical bistability of a driven, interacting Rydberg gas. Its order parameter is the Rydberg population; the light is a probe.

This note keeps their sensing protocol and swaps the phase transition. We use a *light–matter* transition, in which the cavity field itself is the order parameter, because the microwave field then sets the dressed-polariton splitting $\tilde{\omega}_0 = \delta + \Omega_\mu/2$ and therefore *moves the phase boundary.* Two flavours are available. **Route I** is the rotating-wave transition of Zhang *et al.* [PRL **110**, 090402 (2013)]; in a driven cavity it is realised as a superradiant *lasing* threshold, and it needs no new light–matter physics beyond a repump. **Route II** is the $\mathbb{Z}_2$ Dicke transition of Dimer *et al.* / Baden *et al.*, which requires engineered counter-rotating terms and is the natural second step.

Worked out here, with numerics. Route I has a threshold $G^2_{\text{th}} = \kappa\Gamma[1 + D^2/(\kappa+\Gamma)^2]$ (verified against linear stability to 8 digits), and in the bad-cavity limit the emitted frequency locks to the atoms: $\partial\nu/\partial\tilde{\omega}_0 = 0.990$ while $\partial\nu/\partial\omega_{\text{cav}} = 9.9\times10^{-3}$, so **cavity drift is rejected** $101\times$ at $\kappa/\Gamma = 100$. That directly answers the standing objection to critical sensing — that criticality amplifies drift as readily as signal. Reading the measurand as an optical *frequency* rather than an amplitude gives $\sqrt{S_E} = 0.070\ \text{nV}\,\text{cm}^{-1}\text{Hz}^{-1/2}$ at $N = 10^6$, a factor 5.1 better than Route II at equal atom number because the lasing route uses all N atoms whereas Route II is capped by $n_b \ll N$. Route II in exchange offers a genuine ground-state critical point, ε^{-1} transduction gain and 10 dB of self-generated squeezing.

Both routes reach the same $\sqrt{\Gamma/N}$ floor — criticality buys gain and drift immunity, not a new scaling law. Adding Rydberg interactions to Route I bends the threshold back into optical bistability, which shows that the incumbent transition and the proposed one are two limits of a single model, with the superradiant solid as the coexistence region. A further difference from the incumbent is that both routes here keep the resonant Autler–Townes coupling, so the scale factor is exactly $-d/2\hbar$; Refs. [6, 7] couple the field through a *quadratic* AC Stark shift about a nonzero bias, with a fitted scale factor.

Two caveats are load-bearing and stated where they arise. Route I assumes one can invert a ground-to-dressed-Rydberg transition; no superradiant laser has been built on anything but a narrow optical transition, and the alternative architecture costs sensitivity (Sec. 3). For Route II the dominant error is not quantum but technical: the Raman pump light shift adds directly to $\tilde{\omega}_0$, so 10^{-3} intensity noise on a 1 MHz shift gives 0.88 μV/cm of drift, some 5×10^3 times the one-second quantum floor (Sec. 4.3.4).

Contents

1 Same protocol, different phase transition

The proposal is best stated as a substitution rather than an invention.

Ding *et al.* [6] and Wang *et al.* [7] have established a sensing protocol: bias a Rydberg ensemble at a critical point, let the microwave field push the system across it, and read the steep optical response. It works — 49 and 2.6 $\mathrm{nV\,cm^{-1}Hz^{-1/2}}$ respectively. The critical point they use is that of a *non-equilibrium phase transition* (NEPT): the optical bistability of a laser-driven, interacting Rydberg gas, arising from the competition between long-range Rydberg interactions, drive and dissipation. Its order parameter is the Rydberg population, observed in transmission. The light is a probe; delete the cavity and the transition survives.

This note keeps the protocol and changes the transition. We use a **light–matter** transition, in which the cavity field is itself the order parameter — delete the cavity and there is no transition at all. The reason this is a sensor is one equation. In the Rydberg-EIT ladder the microwave field dresses the two Rydberg states and thereby fixes the splitting $\tilde{\omega}_0$ of the effective two-level system the cavity sees,

$$\tilde{\omega}_0 = \delta + \Omega_\mu/2, \qquad \Omega_\mu = dE_\mu/\hbar, \tag{1}$$

and every light–matter phase boundary depends on $\tilde{\omega}_0$. *The boundary is therefore a function of the field to be measured.*

	NEPT (incumbent) Ding 2022, Wang 2025	**Route I**: $U(1)$ after Zhang 2013	**Route II**: $\mathbb{Z}_2$ after Dimer 2007 / Baden 2014
Driven by	Rydberg interactions	light–matter coupling (RWA)	light–matter coupling (+ counter-rotating)
Order parameter	Rydberg population	phase of emitted field	cavity field $\langle a \rangle$
Broken symmetry	none (Ising-like jump)	$U(1)$	$\mathbb{Z}_2$
Realisation	optical bistability	superradiant lasing threshold	ground-state QPT
Extra hardware	—	repump	second Raman channel

Table 1: The substitution. Sections 3 and 4 work out the two new columns; Sec. 6 compares all three.

Two flavours of light–matter transition are available, and they differ in how much new physics they demand.

Route I (Sec. 3) is the transition of Zhang *et al.* [2] — rotating-wave, $U(1)$-breaking. Their calculation is at equilibrium, at fixed chemical potential; a driven optical cavity has no chemical potential, so the same broken symmetry is realised instead as a *superradiant lasing* threshold. This is the cheaper route: it adds only a repump to apparatus that already exists [33]. Its distinctive payoff is that in the bad-cavity limit the emitted frequency locks to the atoms, so the measurand is read as an optical frequency and cavity drift is rejected.

Route II (Sec. 4) is the $\mathbb{Z}_2$ Dicke transition, which needs engineered counter-rotating terms [11, 12, 29]. It is the more demanding experiment and the stronger physics claim: a genuine ground-state critical point, with divergent transduction gain and self-generated squeezing.

Section 2 sets up the shared front end, Secs. 3 and 4 work out the two routes, Sec. 5 puts the Rydberg interactions back in and shows that they bend Route I into the incumbent NEPT, Sec. 6 weighs the three against each other, Sec. 7 reviews what has been done, and Sec. 8 assesses.

A note on sourcing. The primary sources for every claim of the form "paper X does not do Y" in Sec. 7 have been read first-hand: Refs. [1, 3, 4] (abstracts and stated results), and Refs. [2, 6, 7] in full. One substantive correction came out of that reading and is recorded in Sec. 3: the model of Ref. [2] is written in the rotating-wave approximation, so its transition is $U(1)$ rather than the $\mathbb{Z}_2$ one, which is why Route I and Route II are separated here rather than conflated.

2 The model

2.1 Level scheme and the dressed sensor spin

Take the standard Rydberg-EIT electrometry ladder in ^{87}Rb: $|g\rangle = |5S_{1/2}\rangle$, $|e\rangle = |5P_{3/2}\rangle$, $|r\rangle = |53D_{5/2}\rangle$, $|r'\rangle = |54P_{3/2}\rangle$. A probe field addresses $|g\rangle \to |e\rangle$, a strong control laser Ω_c drives $|e\rangle \to |r\rangle$, and the microwave field to be measured drives $|r\rangle \to |r'\rangle$ with Rabi frequency $\Omega_\mu = dE_\mu/\hbar$. For this transition $d = 1774.82\, ea_0$, giving the conversion

$$\Omega_\mu/2\pi = 2.271\ \text{MHz} \times \left(E_\mu/\text{mV cm}^{-1}\right). \tag{2}$$

On microwave resonance the Rydberg pair is dressed into $|\pm\rangle = (|r\rangle \pm |r'\rangle)/\sqrt{2}$ at energies $\pm\Omega_\mu/2$: this is the Autler–Townes doublet that conventional electrometry reads out as a line splitting. The key move here is to treat one dressed state, say $|+\rangle$, together with $|g\rangle$ as a single effective spin-$\frac{1}{2}$ whose splitting is

$$\tilde{\omega}_0 = \delta + \Omega_\mu/2, \tag{3}$$

δ being the two-photon detuning of the optical fields. *The measurand has become a Hamiltonian parameter of a two-level system*, not a spectroscopic feature to be resolved.

2.2 Two ways to make a light–matter transition

A collection of N atoms coupled to a cavity in the rotating-wave approximation,

$$H_{\mathrm{TC}} = \omega\, a^\dagger a + \tilde{\omega}_0\, b^\dagger b + G\,(a^\dagger b + a\, b^\dagger), \qquad G = g_0\sqrt{N}, \tag{4}$$

has *no* phase transition on its own: it is two coupled linear oscillators, and what one observes is the collective Rabi splitting of Refs. [3, 4]. Getting a transition out of light and matter requires adding one of two things.

Add gain (Route I). Incoherently repump the atoms. Equation (4) then acquires a threshold at which the collective gain balances the cavity loss, above which the field acquires a spontaneously chosen phase — a $U(1)$-breaking superradiant lasing transition. This is the driven-dissipative counterpart of the polariton condensation of Ref. [2], whose equilibrium version is reached instead by tuning a chemical potential. Hardware cost: a repump beam.

Add counter-rotating terms (Route II). Restore $a^\dagger b^\dagger + ab$ to Eq. (4) and the model becomes the Dicke model, with a $\mathbb{Z}_2$ symmetry-breaking transition already in the *ground state*.

It is worth being precise about why these terms have to be engineered, because two distinct obstacles are often run together. The counter-rotating terms are *natively present*: the dipole coupling is $g(a+a^\dagger)(\sigma^+ + \sigma^-)$ and contains them identically. The problem is that they oscillate at $\omega_c + \omega_0$ — twice an optical frequency — and average away under the rotating-wave approximation unless $g/\omega_0 \sim 1$. For a 780 nm transition $\omega_0/2\pi = 384$ THz, so $g/\omega_0 = 2.6 \times 10^{-9}$ for a single atom at $g_0/2\pi = 1$ MHz, and still only 2.6×10^{-6} collectively at $N = 10^6$. Put differently: with both frequencies optical the Dicke threshold would sit at $\lambda_c/2\pi = 192$ THz, against a best collective coupling of order 10 GHz — short by four orders of magnitude. *Separately*, even if that gap could be closed, the A^2 term and the Thomas–Reiche–Kuhn sum rule would forbid the superradiant transition for a bare dipole coupling [15, 16].

The engineering circumvents both at once, and it does not conjure the terms from nothing. It places a term of the required *form* into an effective Hamiltonian written in a rotating frame, where ω and $\tilde{\omega}_0$ are detunings of order MHz rather than optical frequencies. The threshold then sits at $\lambda_c/2\pi = 0.5$ MHz for $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$ MHz, which an engineered coupling reaches easily: the change of frame buys a factor $\sim 4 \times 10^8$ in what the threshold demands. The no-go theorem does not constrain the result because the effective couplings are set by drive strengths and detunings, which are free parameters, not by a sum rule on a bare dipole. Reference [2] makes the complementary point that in a two-photon excitation scheme, where $|g\rangle$ is not directly dipole-coupled to the Rydberg state, the standard no-go argument does not apply in the first place.

Concretely, the required term is generated by driving with *two* tones. In the scheme of Dimer, Estienne, Parkins and Carmichael [11], realised by Baden *et al.* [12], one tone drives the Raman process that absorbs a cavity photon while exciting the atom (co-rotating), and a second, higher-frequency tone drives the process that *emits* a cavity photon while exciting the atom — the extra energy coming from that tone. Modulating a single pump is simply a compact way to produce the two tones as sidebands, and this is the route Ma *et al.* [29] take: an explicit periodic-driving recipe for the *anisotropic* Dicke model in a cavity-coupled Rydberg array, with the ratio of counter-rotating to rotating terms tunable from zero to infinity. That is precisely the knob Route II needs, and it is already published. Hardware cost: a second tone, whether from a second laser or from modulating the first.

Without the second tone, nothing happens at all. This is worth stating starkly, because it shows the counter-rotating terms are the mechanism rather than a refinement of it. With the co-rotating coupling alone, Eq. (4) conserves the excitation number $a^\dagger a + b^\dagger b$, and the experimentally natural initial state — every atom in $|g\rangle$, no cavity photons — has zero excitations. It is therefore an exact eigenstate, and the cavity stays dark forever however hard one pumps. We checked this numerically at $N = 8$, $\lambda = 1.2\,\lambda_c$: the residual $\|(H - \langle H\rangle)|g, 0\rangle\|$ is 0 to machine precision for the Tavis–Cummings coupling and $\langle a^\dagger a\rangle$ remains identically zero at all times, whereas restoring the counter-rotating terms gives a residual of 0.6 and $\langle a^\dagger a\rangle$ grows to $\sim$2. Above threshold $|g, 0\rangle$ is not merely non-stationary but *unstable*: the field grows out of vacuum fluctuations, exactly as a laser turns on, and saturates when the collective spin does. Note that "everything radiating" overstates it — near threshold the Rydberg fraction is $p_r = (1 - \mu)/2$, which vanishes linearly as the critical point is approached (Sec. 4.3.4).

In both cases the measurand enters only through $\tilde{\omega}_0$, which is what makes Eq. (1) a sensing curve.

2.3 Which frequency the sensor listens to

Two tuning questions are easy to conflate, and they have different answers.

The carrier is set by the Rydberg pair, exactly as in conventional electrometry. Neither route changes this. Transition frequencies scale as n^{-3}, so the accessible range runs from hundreds of GHz at low n to sub-GHz at high n — Ref. [6] uses $51D_{3/2} \to 52P_{1/2}$ at 16.6 GHz, Ref. [7] $47D_{5/2} \to 48P_{3/2}$, and the worked example here $53D_{5/2} \to 54P_{3/2}$. The available transitions form a *comb*, not a continuum: at $n \approx 50$ the $\Delta n = 1$ teeth are spaced by roughly $3 \times (2\mathrm{Ry}/n^4)$, a few GHz, and other ℓ and Δn densify it without closing the gaps. Filling them is solved technology: Ref. [31] AC-Stark-tunes $78S_{1/2} \to 78P_{1/2}$ with a 70 MHz field applied *outside* the vapour cell and uses Floquet sidebands to reach 1.172 GHz of continuous coverage; DC-Stark tuning is the other standard route [32].

The bias point is set optically, and that is the new knob. Where on the sensing curve the system sits is fixed by the two-photon detuning δ (and, in Route I, the cavity detuning ω_{cav}) — independently of the carrier selection above. Since $\tilde{\omega}_0 = \delta + \Omega_\mu/2$ must sit at a specific small value, δ has to track the microwave amplitude to stay biased: the lock *is* the measurement, which is Protocol B. Refs. [6, 7] make the same statement about their own systems, noting that the sensitive operating field "can be tuned by the coupling detuning".

A tension specific to Route I. Route I depends on $\Gamma \ll \kappa$: the drift rejection *is* κ/Γ and the floor is $\sqrt{2\Gamma/N}$, so broadening Γ costs twice. Rydberg polarizabilities scale as n^7 — of order 10^3 MHz/(V/cm)^2 for Rb nS at $n \approx 50$ — so a 1 GHz DC-Stark shift needs $\sim$ 0.8 V/cm, where the slope $2\alpha E$ is a few MHz per (mV/cm). A field inhomogeneity of even 1 mV/cm across the ensemble then broadens the transition by megahertz, two orders above the 20 kHz Route I assumes. *DC-Stark gap-filling is therefore largely incompatible with Route I*, which should sit on a natural Rydberg resonance, or use the externally applied AC-Stark/Floquet scheme of Ref. [31] where the tuning field is more uniform. Route II is more forgiving: γ enters its floor but there is no κ/Γ rejection factor to protect. (This is an order-of-magnitude argument from the n^7 scaling and should be checked against a proper polarizability calculation.)

3 Route I: the $U(1)$ (lasing) transition

3.1 From Zhang's condensation to a lasing threshold

Reading Ref. [2] in full turns up a point that shapes this whole section. Their Hamiltonian couples the cavity as $(g/\sqrt{N})\sum_i(b_i^\dagger a_i\psi + \text{H.c.})$ — a rotating-wave coupling that conserves the total excitation number $N_{\text{ex}} = \psi^\dagger\psi + \sum_i n_i$, which is why their phase diagram is drawn against a chemical potential $\tilde{\mu}$ conjugate to N_{ex}. Their "generalised Dicke model" is therefore of Tavis–Cummings type, and their superradiance is a $U(1)$-breaking polariton condensation, not the $\mathbb{Z}_2$ transition of Sec. 4.

An optical cavity has no chemical potential; it has pump and loss. The same broken symmetry is therefore realised in a driven system as a *superradiant lasing* threshold. The universality class differs from Zhang's equilibrium condensation — this is a non-equilibrium, gain-driven transition — but the symmetry, the order parameter (the phase of the emitted field) and, crucially for us, the dependence of the boundary on $\tilde{\omega}_0$ all carry over.

3.2 Threshold

Semiclassical Maxwell–Bloch equations for a homogeneously broadened collective transition, in a frame rotating at the emission frequency ν:

$$\dot{a} = -(\kappa + iD_c)\,a + \tfrac{G^2}{g}\,p, \qquad \dot{p} = -(\Gamma + iD_a)\,p + g\,a\,s, \qquad \dot{s} = -\gamma_{\|}(s - s_0) - 2g(a^*p + \text{c.c.}), \quad (5)$$

with $D_c = \omega_{\text{cav}} - \nu$, $D_a = \tilde{\omega}_0 - \nu$, transverse decay Γ (including repump broadening), and collective gain $G^2 = g^2 N s_0/2$. Linearising about $a = p = 0$ gives $G^2 = (\kappa + iD_c)(\Gamma + iD_a)$. The imaginary part fixes the emission frequency and the real part the threshold. Writing $D \equiv \omega_{\text{cav}} - \tilde{\omega}_0$ for the atom–cavity detuning,

$$\nu = \frac{\kappa\,\tilde{\omega}_0 + \Gamma\,\omega_{\text{cav}}}{\kappa + \Gamma} \qquad \text{(cavity pulling)}, \qquad (6)$$

$$G_{\text{th}}^2 = \kappa\Gamma\left[1 + \frac{D^2}{(\kappa + \Gamma)^2}\right] \qquad \text{(threshold)}. \qquad (7)$$

We verified Eq. (7) against the largest real eigenvalue of the linearised (a, p) system: the two agree to all 8 digits printed, at $D/2\pi = 0$, 1, 3 and 8 MHz.

Since $D = \omega_{\text{cav}} - \delta - \Omega_\mu/2$, *the microwave field moves the lasing threshold.* At fixed collective gain the boundary is crossed at $|D_{\text{th}}| = (\kappa+\Gamma)\sqrt{G^2/\kappa\Gamma - 1}$; equivalently, servoing the two-photon detuning to hold the system at threshold gives

$$\delta_{\text{th}} = \omega_{\text{cav}} \mp (\kappa + \Gamma)\sqrt{\tfrac{G^2}{\kappa\Gamma} - 1} - \frac{\Omega_\mu}{2}, \qquad \frac{\partial\delta_{\text{th}}}{\partial E_\mu} = -\frac{d}{2\hbar}, \qquad (8)$$

the same SI-traceable scale factor as Autler–Townes electrometry. Figure 1(a) shows the resulting sensing curve.

3.3 The frequency read-out, and why it answers the drift objection

Equation (6) is the reason to prefer this route. The emission frequency is a weighted mean of the atomic and cavity frequencies, and in the bad-cavity limit $\kappa \gg \Gamma$ the atomic weight dominates:

$$\frac{\partial\nu}{\partial\tilde{\omega}_0} = \frac{\kappa}{\kappa + \Gamma} \to 1, \qquad \frac{\partial\nu}{\partial\omega_{\text{cav}}} = \frac{\Gamma}{\kappa + \Gamma} \to \frac{\Gamma}{\kappa} \ll 1. \qquad (9)$$

At $\kappa/2\pi = 2$ MHz and $\Gamma/2\pi = 20$ kHz we find 0.9901 and 9.90×10^{-3}: the signal is transferred essentially in full while cavity-length noise is suppressed 101×. The measurand appears as

$$\frac{\partial\nu}{\partial E_\mu} = \frac{\kappa}{\kappa + \Gamma}\cdot\frac{d}{2\hbar} = 1.124 \text{ MHz per (mV/cm)}, \qquad (10)$$

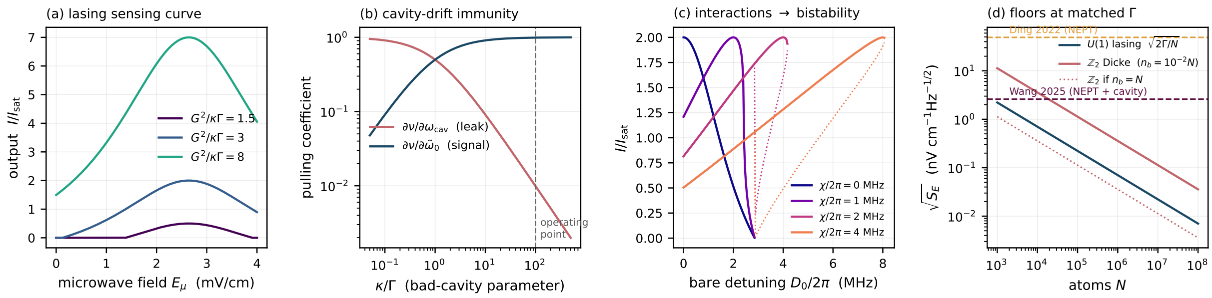

Figure 1: Route I. $\kappa/2\pi = 2$ MHz, $\Gamma/2\pi = 20$ kHz. (a) Output power versus microwave field: the field tunes the atom–cavity detuning through resonance, switching the laser on and off. (b) Cavity-pulling coefficients. In the bad-cavity limit the emission frequency follows the atoms ($\partial\nu/\partial\tilde{\omega}_0 \to 1$) and ignores the cavity ($\partial\nu/\partial\omega_{\rm cav} \to \Gamma/\kappa$): at the operating point cavity drift is rejected 101×. (c) With a Rydberg-interaction shift χ pulling the transition towards resonance, the threshold folds over into bistability (dotted = unstable branch; $I = 0$ remains a solution throughout the folded region) — this is the incumbent NEPT, reached from the lasing side. (d) Quantum-noise floors at matched Γ, against the two published NEPT results.

i.e. 0.990 times the half-Autler–Townes slope — read out as an *optical frequency*, by beating against a reference.

This is the substantive advantage over both the incumbent NEPT and Route II. The standing objection to critical sensing is that a divergent susceptibility amplifies drift as readily as signal; formally, the critical Fisher-information matrix is sloppy [34]. Here the dominant technical channel — cavity length — is not amplified but *actively rejected*, by the same mechanism that makes bad-cavity ("active optical clock") lasers insensitive to their resonators. The rejection is a design parameter: it is κ/Γ, tunable by choosing the cavity finesse.

3.4 Sensitivity floor

Above threshold each repumped atom yields one cavity photon, $2\kappa n = Nw/2$, and the bad-cavity laser linewidth is the Schawlow–Townes value reduced by the squared pulling coefficient,

$$\Delta\omega = \frac{\kappa}{2n}\left(\frac{\Gamma}{\kappa+\Gamma}\right)^2 \xrightarrow[\kappa\gg\Gamma,\ w=\Gamma]{} \frac{2\Gamma}{N}. \tag{11}$$

Phase diffusion at rate $\Delta\omega$ gives $\sqrt{S_\nu} = \sqrt{\Delta\omega}$, so referring through the pulling coefficient and through $\tilde{\omega}_0 = \delta + \Omega_\mu/2$,

$$\boxed{\sqrt{S_{\tilde{\omega}_0}} = \frac{\kappa+\Gamma}{\kappa}\sqrt{\frac{2\Gamma}{N}}} \tag{12}$$

— the collective-atomic standard quantum limit, with all N atoms participating. Note this is a derivation of the floor, not an independent check of it: the two expressions are the same formula. In physical units:

An unexamined assumption, and the design fork it opens. Everything above assumes the lasing transition is $|g\rangle \leftrightarrow |+\rangle$, i.e. that one can *invert* a ground-to-dressed-Rydberg transition and hold it inverted. This note does not establish that, and it is not a small matter. At $w = \Gamma = 2\pi \times 20$ kHz each atom cycles every $\sim 8\,\mu$s, comfortably inside a $\sim 100\,\mu$s Rydberg lifetime, so it is not excluded — but it means driving a large fraction of the ensemble into Rydberg states continuously, with the blockade, ionisation and collisional problems that brings, and in direct tension with the narrow Γ the whole route depends on. Every superradiant laser actually built uses a narrow *optical* transition (Sr $^3P_0$), not a Rydberg state.

N	intracavity photons	linewidth	$\sqrt{S_E}$ (nV cm^{-1}Hz$^{-1/2}$)	Route II at $n_b = 10^{-2}N$
10^4	25	3.9 Hz	0.703	3.58
10^5	250	392 mHz	0.222	1.13
10^6	2.5×10^3	39 mHz	0.070	0.358
10^7	2.5×10^4	3.9 mHz	0.022	0.113
10^8	2.5×10^5	0.39 mHz	0.0070	0.036

Table 2: Route I frequency-pulling read-out, $\kappa/2\pi = 2$ MHz, $\Gamma/2\pi = 20$ kHz, repump $w = \Gamma$. The last column is Route II's floor at the same N and Γ (Sec. 4), which is capped by the Holstein–Primakoff requirement $n_b \ll N$.

There is a plausible alternative architecture: let the lasing transition be a narrow optical one and have the Rydberg admixture *shift* it, so the microwave field enters as a perturbation on a clock transition rather than as the lasing transition itself. That preserves the bad-cavity physics and the drift rejection, but the microwave sensitivity is then suppressed by the dressing fraction, which would degrade the table above by that factor. Which architecture to pursue is an open design decision, and the numbers quoted here belong to the first one. Route II carries no analogous assumption: it requires no inversion at all, since its normal phase is simply every atom in $|g\rangle$.

Subject to that, Route I is a factor 5.1 better at equal atom number, for a structural reason worth stating plainly: the lasing route inverts and uses *all* N atoms, whereas Route II linearises about the unexcited state and is limited to $n_b \ll N$ collective excitations. The mHz linewidths are the standard superradiant-laser prediction and are the reason the frequency read-out is competitive at all.

4 Route II: the $\mathbb{Z}_2$ Dicke transition

Restoring the counter-rotating terms to Eq. (4) turns the Tavis–Cummings model into the Dicke model,

$$H = \omega\, a^\dagger a \;+\; \tilde{\omega}_0 J_z \;+\; \frac{2\lambda}{\sqrt{N}}\,(a + a^\dagger)\, J_x, \tag{13}$$

with J_α the collective spin operators in the $j = N/2$ manifold and λ the engineered coupling of Sec. 2.2. Two features matter for sensing. First, ω and $\tilde{\omega}_0$ are *effective* frequencies — detunings in a rotating frame — so they can be dialled from kHz to MHz and reaching $\lambda \sim \sqrt{\omega\tilde{\omega}_0}/2$ does not require ultrastrong coupling. Second, $\lambda \propto \Omega_{\mathrm{Raman}} g_0 \sqrt{N}/\Delta_e$ is set by laser power and is tunable *in situ*, so the operating point can be servoed onto threshold.

4.1 Threshold line and order parameter

Mean-field minimisation of Eq. (13) reproduces the textbook result [14]. Writing $\langle J_z\rangle = -\frac{N}{2}\cos\theta$, $\langle J_x\rangle = \frac{N}{2}\sin\theta$ and $\alpha = \langle a\rangle/\sqrt{N}$,

$$\frac{E}{N} = \omega\alpha^2 - \frac{\tilde{\omega}_0}{2}\cos\theta + 2\lambda\alpha\sin\theta, \qquad \alpha = -\frac{\lambda}{\omega}\sin\theta, \tag{14}$$

which has the normal solution $\theta = 0$ for $\lambda < \lambda_c$ and, for $\lambda > \lambda_c$,

$$\cos\theta = \mu \equiv \frac{\lambda_c^2}{\lambda^2} = \frac{\omega\tilde{\omega}_0}{4\lambda^2}, \qquad \bar{n} \equiv \frac{\langle a^\dagger a\rangle}{N} = \frac{\lambda^2}{\omega^2}\left(1 - \mu^2\right), \qquad \lambda_c = \tfrac{1}{2}\sqrt{\omega\tilde{\omega}_0}. \tag{15}$$

Figure 2 shows the resulting sensing curve and the order parameter as the microwave field is swept at fixed λ.

The immediate observation is that the *sign* of the response is the useful one: for $\delta = 0$ a *larger* microwave field raises $\tilde{\omega}_0$ and hence λ_c, extinguishing superradiance. The sensor is therefore naturally operated as a bright-to-dark discriminator, which is favourable for detection (one measures a decrease from a large signal, not an increase from zero).

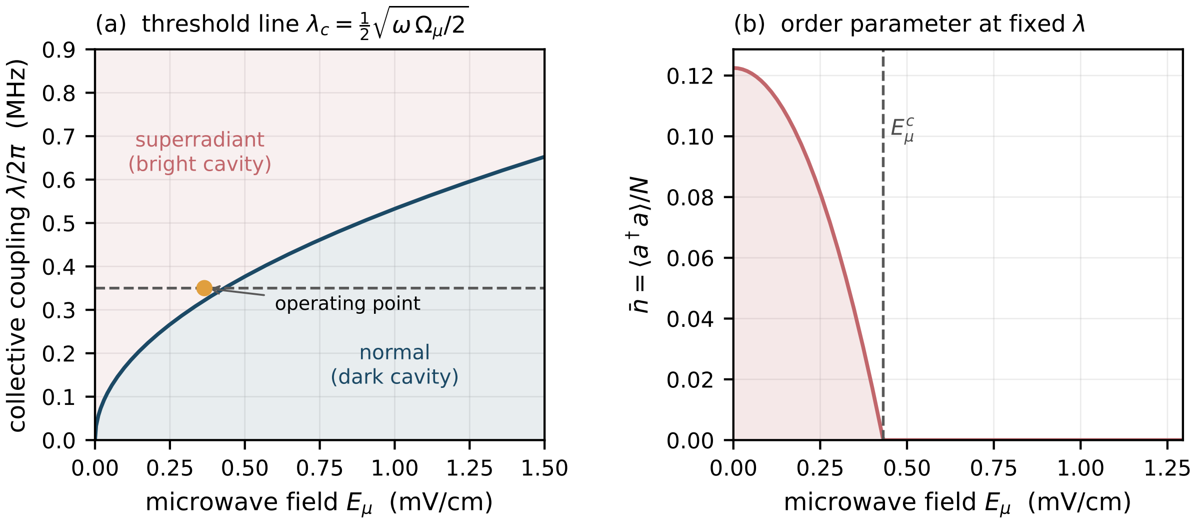

Figure 2: The microwave field moves the phase boundary. (a) Threshold coupling $\lambda_c = \frac{1}{2}\sqrt{\omega\,\Omega_\mu/2}$ versus microwave field for $\omega/2\pi = 1$ MHz, $\delta = 0$ and the $53D_{5/2} \to 54P_{3/2}$ dipole. At fixed λ (dashed) the system crosses from superradiant to normal at a well-defined field. (b) Order parameter $\bar{n}$ at $\lambda/2\pi = 0.35$ MHz: the cavity goes dark above a critical field E_μ^c. The steep square-root onset near E_μ^c is the transduction gain the scheme trades on.

4.2 Three read-out protocols

The construction supports three distinct protocols, with quite different characters. It is worth separating them because most of the confusion in the critical-metrology literature comes from conflating them.

Protocol A — order-parameter read-out (analogue). Bias at fixed λ slightly above threshold and monitor the emitted light. The signal is $\partial\alpha/\partial\tilde{\omega}_0$, which diverges as $\mu \to 1$:

$$\frac{\partial\alpha}{\partial\tilde{\omega}_0} = \frac{\mu}{4\lambda\sqrt{1-\mu^2}} \xrightarrow[\mu\to 1]{} \infty. \tag{16}$$

This is the direct analogue of what Ding *et al.* [6] and Wang *et al.* [7] do with the Rydberg-interaction-driven critical point, and it is the protocol most likely to be implemented first. Its weakness is that the divergent susceptibility is not selective: laser intensity noise (which moves λ), cavity length drift (which moves ω) and stray dc fields (which move δ) are amplified by exactly the same factor as the signal.

Protocol B — threshold (null) read-out. Hold λ and ω fixed and servo the two-photon detuning δ so that the system sits exactly at threshold. The lock point satisfies

$$\delta_{\rm th} = \frac{4\lambda^2}{\omega} - \frac{\Omega_\mu}{2} \qquad \Longrightarrow \qquad \frac{\partial\delta_{\rm th}}{\partial E_\mu} = -\frac{d}{2\hbar}, \tag{17}$$

so a change in the microwave field is read out directly as a change in an *optical laser detuning*, which is a frequency and is therefore SI-traceable in exactly the sense that makes Autler–Townes electrometry attractive [9, 10]. The scale factor $d/2\hbar$ is the same as for Autler–Townes; what changes is the shape of the discriminant. In AT electrometry the measurand is a line *splitting*, fitted against the EIT linewidth; here it is the position of an *edge*, held by a servo.

Figure 3 shows the apparatus. The observable is worth stating explicitly because it is easy to assume otherwise: it is *not* EIT transparency. There is no probe on the sensing transition. The cavity is dark below threshold and *emits* above it, and one detects the light leaking through the

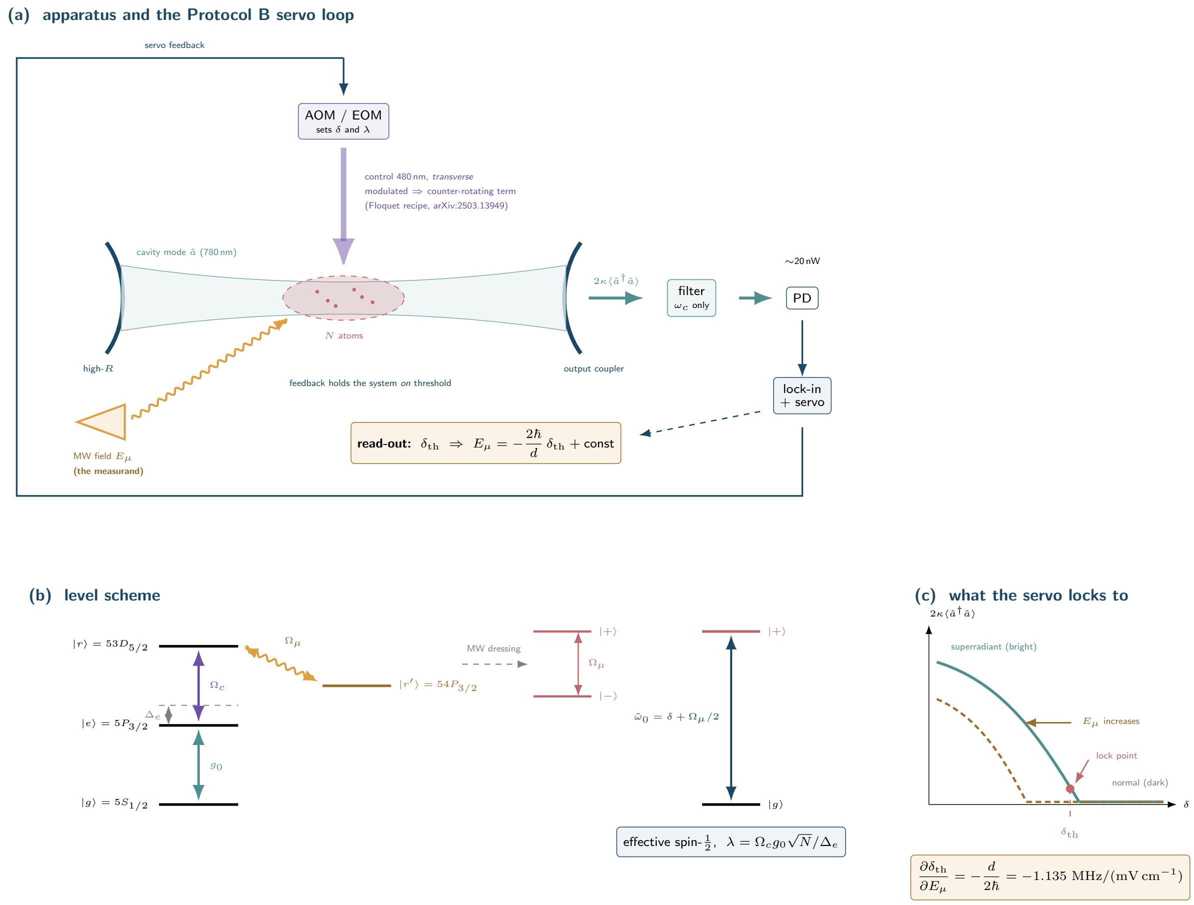

Figure 3: Route II with the Protocol B read-out. (a) The cavity mode is one leg of the Raman process, so the pump is applied *transversely* and never enters the mode; the counter-rotating term is generated by modulating it, following the recipe of Ref. [29]. The microwave field to be measured dresses the Rydberg pair. Cavity emission is filtered, detected, and fed back onto the two-photon detuning; the servo's correction *is* the measurement. Pump rejection is geometric (transverse pumping, ~60–80 dB) plus the cavity's own filtering (~70 dB at finesse 10^4), against a requirement of ~40–60 dB. (b) The microwave field dresses $|r\rangle, |r'\rangle$ into $|\pm\rangle$; $|g\rangle$ and $|+\rangle$ form the effective spin whose splitting $\tilde{\omega}_0 = \delta + \Omega_\mu/2$ carries the measurand. (c) The servo holds the system on the edge, which translates with the field.

output coupler, $2\kappa\langle a^\dagger a\rangle$ — of order 20 nW at $N = 10^6$ (Sec. 4.3.4). Figure 4 shows the computed trace.

An honest limit on this protocol. It is tempting to argue that a threshold must be a sharper discriminant than a resonance, and an earlier draft of this note did. The numerics do not support it. From Fig. 4(b) the edge is rounded by finite N only over $\sim \tilde{\omega}_{0\mathrm{th}} N^{-2/3}$, which is $\sim$ 50 Hz at $N = 10^6$ and therefore negligible; the width is set instead by the atomic decoherence γ — the same scale that sets the EIT linewidth. Moreover $\partial\bar{n}/\partial\delta$ is *finite* at threshold ($\bar{n}$ rises linearly below it), so the intensity read-out has no divergent slope; the divergence lives in the field amplitude $\alpha \propto \sqrt{\delta_{\rm th} - \delta}$, which is Protocol A's homodyne observable, not this one. Figure 4(d) makes the point visually: conventional EIT-AT responds perfectly well at the same field values.

What survives is narrower but still worth having. Protocol B is a *null* method, so it is first-order insensitive to gain calibration and detector drift — precisely the weakness of Protocol A — and it is frequency-referenced with an exact, dipole-traceable scale factor. It is not demonstrably a sharper discriminant than Autler–Townes, and establishing whether it can be requires the threshold-location noise calculation listed in Sec. 8. To our reading of the literature the null variant has not been proposed for this system.

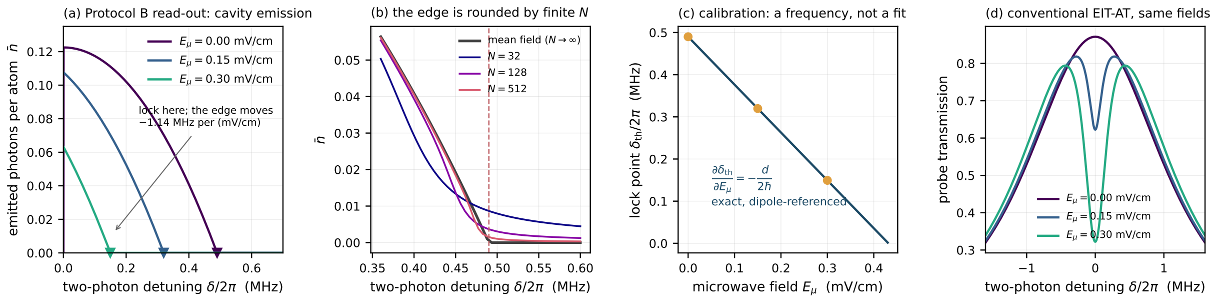

Figure 4: The Protocol B read-out, computed ($\omega/2\pi = 1$ MHz, $\lambda/2\pi = 0.35$ MHz). (a) Emitted photons per atom versus two-photon detuning at three microwave field values; the edge translates at -1.135 MHz per (mV/cm). (b) Exact diagonalisation shows the edge rounded by finite N, following $N^{-2/3}$ — negligible for realistic atom numbers. (c) The calibration line: a frequency, referenced to a known dipole, not a fitted gain. (d) Conventional EIT-AT transmission at the same fields, for contrast. It responds well; the difference between the two read-outs is the *shape* of the discriminant, not that one of them fails.

Protocol C — critical superheterodyne (recommended). Apply a microwave local oscillator with Rabi frequency Ω_{LO} to set the bias point $\tilde{\omega}_0 = \delta + \Omega_{\mathrm{LO}}/2$ just below threshold, and let the weak signal field at $\omega_{\mathrm{LO}} + \omega_{\mathrm{IF}}$ beat against it:

$$\tilde{\omega}_0(t) = \delta + \frac{\Omega_{\mathrm{LO}}}{2} + \frac{\Omega_{\mathrm{s}}}{2}\cos(\omega_{\mathrm{IF}} t). \tag{18}$$

The measurand is now a small *modulation* of a Hamiltonian parameter, amplified by the near-threshold susceptibility and detected at the intermediate frequency. This inherits the linear-in-E_{s} scaling and technical-noise rejection that make the Rydberg superheterodyne receiver [8] the most sensitive Rydberg electrometer built, and adds critical gain on top. It is the configuration used for all the open-system numbers in Sec. 4.3.3, and it is the one we would actually build.

4.3 Metrological analysis

4.3.1 Closed system: quantum Fisher information at criticality

We first ask what the *ground state* of Eq. (13) is worth as a probe of $\tilde{\omega}_0$. Exact diagonalisation in the $j = N/2$ manifold (even-parity sector; photon cutoff converged to 10^{-12}, see Appendix A) gives the quantum Fisher information from the fidelity susceptibility, $F_Q = 8\left(1 - |\langle\psi(\tilde{\omega}_0)|\psi(\tilde{\omega}_0 + h)\rangle|\right)/h^2$.

Figure 5 confirms the expected fully-connected-model exponents [18]: the gap closes as $N^{-1/3}$ and

$$F_Q[\tilde{\omega}_0]\big|_{\lambda=\lambda_c} \propto N^{4/3}, \tag{19}$$

with a fitted exponent of 1.307 (asymptotic value $4/3 = 1.333$; the residual is the expected slow finite-size correction, the fit drifting monotonically upward as the window moves to larger N). In the normal phase we find, at $N = 512$ and in the window $N^{-2/3} \ll \varepsilon \ll 1$,

$$F_Q[\tilde{\omega}_0] \propto \varepsilon^{-2}, \qquad \varepsilon \equiv 1 - \lambda^2/\lambda_c^2, \tag{20}$$

with a fitted exponent of -2.0008 — the Gaussian critical scaling of a squeezed soft mode, cut off at $\varepsilon_{\min} \sim N^{-2/3}$, which is exactly what produces the $N^{4/3}$ peak.

4.3.2 Why the $N^{4/3}$ is not free

Taken at face value, $F_Q \propto N^{4/3}$ says $\delta\Omega_\mu \propto N^{-2/3}$, beating the $N^{-1/2}$ standard quantum limit. This is the claim that critical-metrology proposals are often built on, and it does not survive

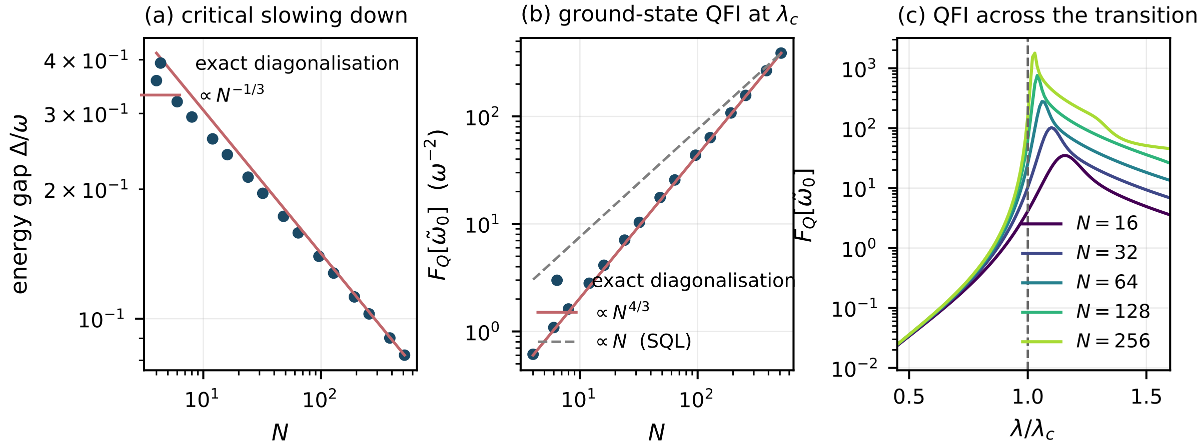

Figure 5: Finite-size scaling at the critical point ($\omega = \tilde{\omega}_0 = 1$, $\lambda = \lambda_c$, $N \leq 512$). (a) The energy gap closes as $N^{-1/3}$ (fitted -0.317 over $N \geq 128$). (b) The quantum Fisher information for $\tilde{\omega}_0$ grows as $N^{4/3}$ (fitted 1.307 over $N \geq 128$), super-extensive and thus formally beyond the standard quantum limit $F_Q \propto N$ (dashed). (c) F_Q across the transition; the peak sharpens and moves towards λ_c as N grows.

proper resource accounting. The gap closes as $N^{-1/3}$, so adiabatic preparation of the critical ground state takes a time $T_{\text{prep}} \propto N^{1/3}$. Converting to a per-root-Hertz sensitivity,

$$\delta\Omega_\mu \sqrt{T_{\text{prep}}} \;\propto\; N^{-2/3} \cdot N^{1/6} \;=\; N^{-1/2}, \tag{21}$$

and the advantage cancels exactly. This is the content of Rams *et al.* [19]; Gietka *et al.* [20] show the cancellation survives shortcuts to adiabaticity, and Gyhm *et al.* [22] have recently established a general T^6 ceiling on the quantum Fisher information of monotonic critical protocols. A proposal built on the $N^{4/3}$ scaling alone would be building on sand, and this document should not be used to write one.

The interesting question is therefore not the scaling but the *prefactor*, which is what the driven open system delivers.

4.3.3 Open system: gain, bandwidth, squeezing and the input-referred floor

We linearise Eq. (13) in the normal phase (Holstein–Primakoff bosons a for the cavity, b for the collective atomic mode), add cavity decay κ and collective decoherence γ, and drive the cavity coherently. In quadratures $v = (x_a, p_a, x_b, p_b)$,

$$\dot{v} = Av + f + Bu, \qquad A = \begin{pmatrix} -\kappa & \omega & 0 & 0 \\ -\omega & -\kappa & -2\lambda & 0 \\ 0 & 0 & -\gamma & \tilde{\omega}_0 \\ -2\lambda & 0 & -\tilde{\omega}_0 & -\gamma \end{pmatrix}, \tag{22}$$

with vacuum inputs u. The measurand enters as $\delta H = \delta\tilde{\omega}_0\, b^\dagger b$, i.e. as a coherent force

$$s \;=\; \delta\tilde{\omega}_0\, \big(0,\, 0,\, \bar{p}_b,\, -\bar{x}_b\big) \tag{23}$$

proportional to the steady-state collective amplitude — which is why Protocol C needs the probe drive. Output homodyne spectra follow from $x_{\text{out}} = \sqrt{2\kappa}\, x_a - x_{\text{in}}$. The implementation is validated by reproducing the exact vacuum level $S_{\text{out}} = 1/2$ for an empty cavity.

Throughout we use $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$ MHz, $\kappa/2\pi = 50$ kHz and $\gamma/2\pi = 5$ kHz, giving a damped threshold $\lambda_c/2\pi = 0.5006$ MHz.

Figure 6 summarises the behaviour. Three points are worth stating precisely.

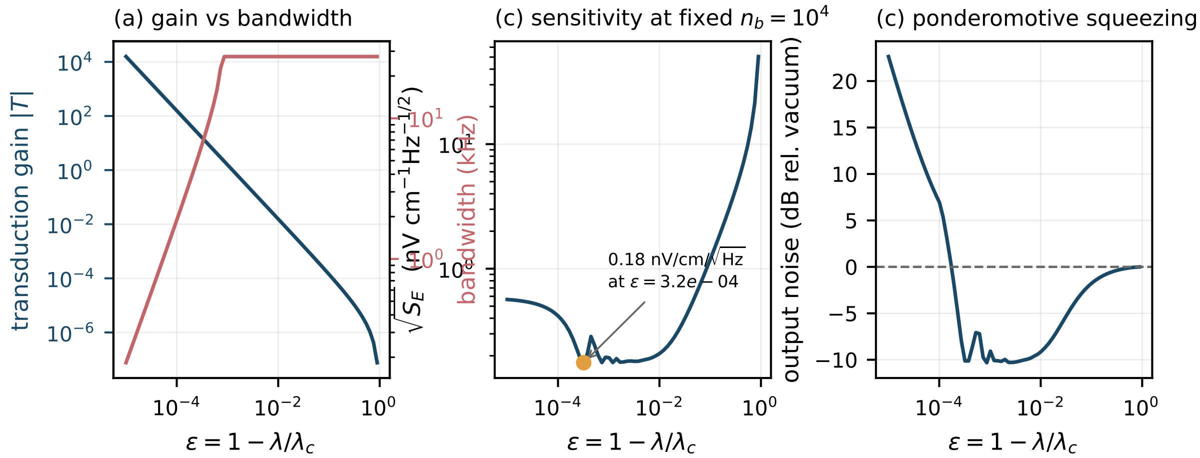

Figure 6: Driven, damped sensor near threshold. (a) Transduction gain (left axis) and response bandwidth (right axis) versus $\varepsilon = 1-\lambda/\lambda_c$. At fixed collective occupation the gain grows as ε^{-1} throughout, while the bandwidth stays pinned at $(\kappa+\gamma)/2 = 27.5$ kHz until $\varepsilon \lesssim 3\times10^{-4}$ and only then collapses as ε^{+1}. There is thus a window of *free* gain before critical slowing down begins. (b) Quantum-noise-limited field sensitivity at fixed $n_b = 10^4$, with a broad optimum of $0.18\ \text{nV cm}^{-1}\text{Hz}^{-1/2}$. (c) The near-threshold system squeezes its own output by up to 10.3 dB before atomic noise takes over.

Gain and bandwidth. At fixed collective occupation $n_b = \langle b^\dagger b\rangle$, the transduction gain scales as $\varepsilon^{-1.000}$ over six decades. The bandwidth, however, is *not* immediately paid for: the slowest relaxation rate stays pinned at $(\kappa+\gamma)/2 = 27.5$ kHz until $\varepsilon \lesssim 3\times10^{-4}$, below which the soft mode becomes overdamped and the rate falls as $\varepsilon^{+1.002}$, giving a strict gain–bandwidth product (constant to 0.78%). Practically: there is roughly three decades of ε over which criticality buys 10^3 in gain at no bandwidth cost, and beyond that it is a straight trade. This is a genuinely useful design statement and it does not appear to have been made for this system.

Intrinsic squeezing. Near threshold the soft mode is a degenerate parametric amplifier and the output is squeezed — we find up to -10.3 dB at $\varepsilon \approx 9\times10^{-4}$, degrading to excess noise once $\varepsilon \lesssim 10^{-4}$ where atomic decoherence dominates. The sensor generates its own squeezed light; no external squeezed source is needed. This is a real, if modest, quantum-enhancement channel and it is compatible with Protocol C.

The control experiment that settles the matter. Sensitivity figures near a critical point are easy to inflate, because approaching threshold increases the steady-state excitation as well as the gain. We therefore repeated the calculation holding n_b *fixed* by re-solving for the drive at each ε. The result is unambiguous:

$$\boxed{\sqrt{S_{\tilde\omega_0}} \;=\; 0.72\,\sqrt{\gamma/n_b}} \tag{24}$$

with the numerical prefactor constant to one part in 10^4 across five decades of n_b and, away from the extreme critical region, across three decades of ε. Equation (24) is the standard quantum limit for n_b atoms with coherence time $1/\gamma$, up to a factor of order unity that we attribute to the intrinsic squeezing.

The physical reading: **criticality is a transducer, not a source of metrological advantage**. It converts the atomic signal into a large optical signal — which is exactly what one needs to beat photodetector and technical noise, and is why Refs. [6, 7] report large Fisher-information gains — but it does not lower the floor set by how many atoms are coherently participating and for how long.

4.3.4 Photon budget, blockade, and the dominant systematic

Asking where the emitted photons come from settles three practical questions.

The light is generated, not transmitted. Each cavity photon is produced by a coherent Raman conversion: a transverse pump photon is absorbed, the collective spin flips, and a photon appears in the cavity mode, $\hbar\omega_{\text{pump}} = \hbar\omega_c + \hbar\Delta_{\text{spin}}$. The intermediate state $|e\rangle$ is far detuned and essentially never populated — this is stimulated, not absorption followed by spontaneous re-emission, and residual real scattering off $|e\rangle$ is precisely the loss that sets γ. Because the cavity mode is one leg of the Raman process, the pump must be applied *transversely* and never enters the mode; discrimination is therefore geometric first ($\sim$60–80 dB) and spectral second (the cavity passes only ω_c, worth $20\log_{10}(\mathcal{F}/\pi) \approx 70$ dB at finesse 10^4), against a requirement of 37 dB for a 0.1 mW pump and 57 dB for 10 mW. The budget closes comfortably.

Blockade is not a limitation at the operating point. The superradiant phase carries a Rydberg population $p_r = (1-\mu)/2$ which vanishes *linearly* as threshold is approached. For a mode volume of $7.9 \times 10^7\,\mu\text{m}^3$ (waist $100\,\mu$m, length 1 cm) holding $N = 10^6$ atoms, and a representative blockade radius $R_b = 6\,\mu$m:

$1-\mu$	10^{-4}	10^{-3}	10^{-2}	10^{-1}	0.5
Rydberg atoms	50	500	5 000	5×10^4	2.5×10^5
mean spacing	$116\,\mu$m	$54\,\mu$m	$25\,\mu$m	$11.6\,\mu$m	$6.8\,\mu$m
in units of R_b	19	9.0	4.2	1.9	1.1
output power	3.9 pW	39 pW	390 pW	3.7 nW	—

Blockade saturation sits at $p_r \approx 0.087$, a factor $\sim$17 further from threshold in $1-\mu$ than any sensible bias point. *This removes a caveat.* The warning in Sec. 5 that blockade caps N_{eff}, and the cavity-occupation suppression reported by Ref. [28], apply deep in the superradiant phase where that study lives — not at a sensing bias point, where the Rydberg gas is dilute. The photon yield is $8\kappa(\lambda/\omega)^2/\Gamma_r \approx 31$ cavity photons per Rydberg decay, independent of $1-\mu$.

The last row of the table is a second, softer constraint the gain–bandwidth analysis does not capture: the light runs out. Detection stays comfortable down to $1-\mu \sim 10^{-4}$ (3.9 pW against a photodiode NEP of order $\text{pW}/\sqrt{\text{Hz}}$), and that is where the operating point should stop, independently of critical slowing down.

The dominant systematic is the pump light shift. The Raman pumps AC-Stark-shift both $|g\rangle$ and $|+\rangle$, and that shift adds *directly* to $\tilde{\omega}_0$ — the same channel as the measurand. Pump intensity noise is therefore indistinguishable from signal. For a differential shift of 1 MHz:

relative intensity noise	10^{-2}	10^{-3}	10^{-4}
equivalent field error	8.8 μV/cm	0.88 μV/cm	0.088 μV/cm

Against a quantum floor of $0.18\ \text{nV}\,\text{cm}^{-1}\text{Hz}^{-1/2}$, a 1 MHz shift with 10^{-3} intensity noise is some 5×10^3 times the one-second floor — and it is drift, so it does not average down. This is the "criticality amplifies drift" objection in its concrete form for Route II, and it is sharper than the generic statement in Sec. 8. Mitigation is mandatory rather than optional: Protocol C's intermediate-frequency encoding to move the signal away from where the drift lives, intensity stabilisation at the 10^{-4}–10^{-5} level, or a choice of Δ_e for which the shifts on $|g\rangle$ and $|+\rangle$ cancel. It is worth noting that Route I answers this objection *physically*, through cavity pulling; Route II must answer it technically.

4.3.5 Sensitivity in physical units

Converting Eq. (24) through $\delta\tilde{\omega}_0 = \delta\Omega_\mu/2$ and $E_\mu = \hbar\Omega_\mu/d$:

Collective occupation n_b	$\sqrt{S_E}$ (nV cm^{-1}Hz$^{-1/2}$)	SQL for $N = n_b$	ratio
10^3	0.567	0.786	0.72
10^4	0.179	0.248	0.72
10^5	0.057	0.079	0.72
10^6	0.018	0.025	0.72
10^7	0.0057	0.0079	0.72

Table 3: Quantum-noise-limited field sensitivity at $\varepsilon = 3\times10^{-3}$, $\gamma/2\pi = 5$ kHz. For reference: the Rydberg superheterodyne receiver of Ref. [8] achieves 55 nV cm^{-1}Hz$^{-1/2}$; Ref. [7] quotes 5.1 nV cm^{-1}Hz$^{-1/2}$ as the best sensitivity from conventional SNR-enhancement methods, 49–50.3 nV cm^{-1}Hz$^{-1/2}$ for free-space critical (non-equilibrium) electrometry, and 2.6 nV cm^{-1}Hz$^{-1/2}$ for its own cavity-enhanced critical measurement.

Two caveats attach to every number in that table. First, Holstein–Primakoff validity requires $n_b \ll N$; taking $n_b/N \lesssim 10^{-2}$, the $n_b = 10^6$ row needs $N \gtrsim 10^8$ atoms coherently coupled to one cavity mode, which is demanding but not absurd for a vapour cell and out of reach for a small cold cloud. Second, these are *quantum-noise-limited* figures with no allowance for Doppler broadening, transit-time effects, inhomogeneous coupling across the cavity mode, laser frequency noise or dc-field drift, all of which are what actually limits real Rydberg electrometers. Treat them as an upper bound on what the mechanism can offer, not a prediction.

5 Rydberg interactions: the link to the superradiant solid

Reference [2] is not the plain Dicke model: its content is what happens when Rydberg–Rydberg interactions compete with the cavity coupling, producing a superradiant *solid* in which superradiance and crystalline order coexist, with first-order boundaries. It is worth asking whether those interactions help or hurt the sensor.

Adding a nearest-neighbour Rydberg repulsion to Eq. (13),

$$H_{\mathrm{DI}} = H + V\sum_{\langle ij\rangle} n_i n_j, \qquad n_i = S_i^z + \tfrac{1}{2}, \tag{25}$$

and taking a two-sublattice mean field on a bipartite lattice of coordination z (with α eliminated by its own stationarity condition),

$$\frac{E}{N} = \omega\alpha^2 - \frac{\tilde{\omega}_0}{4}(\cos\theta_A + \cos\theta_B) + \lambda\alpha(\sin\theta_A + \sin\theta_B) + \frac{Vz}{2}n_A n_B, \qquad \alpha = -\frac{\lambda}{2\omega}(\sin\theta_A + \sin\theta_B), \tag{26}$$

reproduces the phase structure of Ref. [2]. We minimise this from 13×13 seeds per point with L-BFGS-B, and use adiabatic up/down sweeps in λ as the first-order diagnostic. Figure 7 shows the result, and it splits cleanly into two regimes.

At the sensor's operating point, interactions are inert. For $\tilde{\omega}_0 > 0$ the normal-phase Rydberg population vanishes, so $n_A n_B \to 0$ and the interaction term does nothing *at* the transition. On a fine scan we find the onset at $\lambda/\omega = 0.5005$ for every $V \in \{0, 0.5, 2, 5\}$, i.e. no shift resolvable at the 5×10^{-4} level. Only the amplitude above threshold is suppressed (Fig. 7b). This is a genuinely useful robustness result: *the calibration curve Eq.* (1) *is immune to Rydberg interactions*, so atomic density fluctuations — which are exactly what moves the critical point in Refs. [6, 7] — do not move it here. It is the sharpest technical argument for preferring the Dicke critical point over the interaction-driven one.

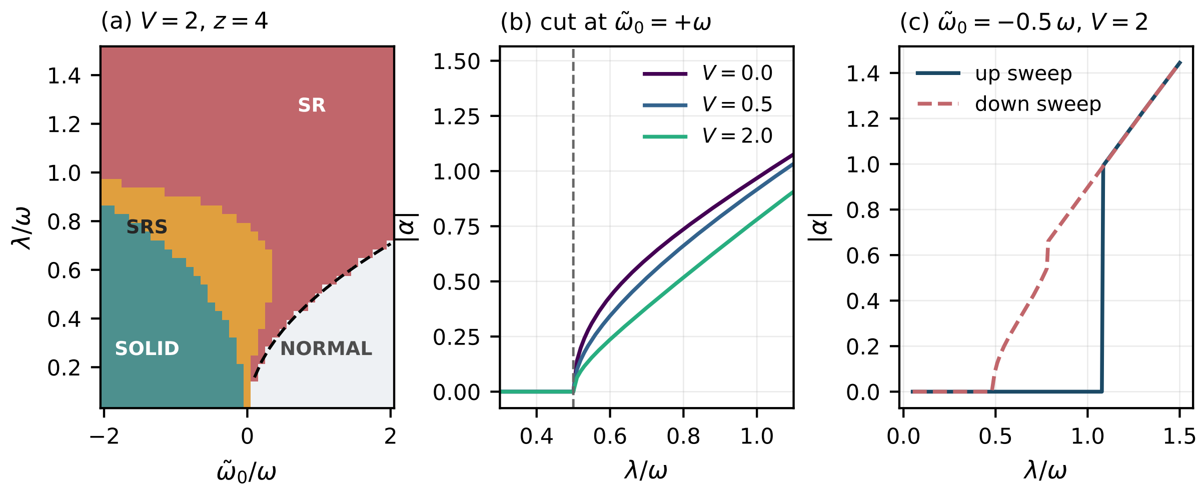

Figure 7: Rydberg interactions and the sensing threshold ($V = 2\omega$, $z = 4$). (a) Mean-field phase diagram. The superradiant solid (SRS) of Ref. [2] occupies only $\tilde{\omega}_0 < 0$; for $\tilde{\omega}_0 > 0$ — the sensor's operating regime — the boundary is the plain Dicke line $\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}$ (dashed), unmodified. (b) Cuts at $\tilde{\omega}_0 = +\omega$ for $V = 0$, 0.5, 2: the onset does not move, only the post-threshold amplitude is suppressed. (c) At $\tilde{\omega}_0 = -0.5\,\omega$ the transition is strongly first order, with a hysteresis loop of height 0.985 in $|\alpha|$.

The superradiant solid lives elsewhere, and it is first order. The SOLID and SRS phases appear only for $\tilde{\omega}_0 < 0$, i.e. when the dressed Rydberg level sits *below* $|g\rangle$ in the rotating frame (readily arranged by choosing $\delta < -\Omega_\mu/2$). There the sequence on increasing λ is SOLID $\to$ SRS $\to$ SR, and the transition is strongly first order: up- and down-sweeps differ by 0.985 in $|\alpha|$ at $\tilde{\omega}_0 = -0.5\,\omega$, $V = 2\omega$.

The design conclusion, however, is largely independent of the details and is worth stating now, because it runs against the intuition that "more Rydberg physics is better":

- **For Protocols A and C, Rydberg interactions are a nuisance.** The clean collective enhancement of the Dicke model assumes N independent emitters sharing one bright mode. Rydberg blockade caps the number of independently excitable atoms per blockade volume at one, so the effective N entering $\lambda = g_0\sqrt{N}$ and the floor (24) is $N_{\text{eff}} = V_{\text{mode}}/V_{\text{blockade}}$, not the atom number. Interactions also dephase the collective mode, raising γ. The quantum Monte Carlo study of Ref. [28] reports exactly that, finding that Rydberg blockade significantly suppresses the cavity occupation. *However*, Sec. 4.3.4 shows this bites only deep in the superradiant phase: at a sensing bias point the Rydberg population vanishes linearly with the distance to threshold, leaving the gas dilute enough that blockade is irrelevant. The tension below is therefore much weaker than it first appears. The sensor wants *weak* interactions — moderate n, or a Rydberg pair chosen for small van der Waals coefficient.
- **For a threshold detector, they are a feature.** If the transition is driven first order, the order parameter jumps discontinuously and the system latches, with hysteresis. That is useless for analogue metrology but excellent for a *trigger*: a device that fires when the field crosses a settable level, with essentially unbounded gain at the switching point and built-in hysteresis to suppress chatter. This is a legitimate secondary application and it is the one place where Ref. [2]'s first-order physics is directly useful.

5.1 Interactions bend Route I back into the incumbent transition

The same question can be asked of Route I, and the answer there is more interesting. Rydberg interactions shift the dressed transition in proportion to the Rydberg population, i.e. to the intracavity intensity, $D(I) = D_0 - \chi I$. Above threshold the gain clamps, so the steady state satisfies

$$I = \frac{G^2}{G_{\rm th}^2(D_0 - \chi I)} - 1. \tag{27}$$

When the shift pulls the transition *towards* resonance the feedback is positive, and Eq. (27) acquires multiple roots. Solving it on a grid at $G^2 = 3\kappa\Gamma$ (so that the bare system lases only for $|D|/2\pi < 2.86$ MHz), we find a single-valued response for $\chi/2\pi \leq 1$ MHz and bistability — two lasing roots coexisting with the dark state $I = 0$ — for $\chi/2\pi \gtrsim 2$ MHz; see Fig. 1(c). At $\chi/2\pi = 2$ MHz and $D_0/2\pi = 4$ MHz the roots sit at $I = 1.61$ and 2.00.

That bistability, with its dark and bright branches and its hysteresis, *is* the non-equilibrium transition Ding *et al.* and Wang *et al.* use. So the incumbent transition and Route I are not rivals from different worlds: they are two limits of one model. Interactions dominant ⇒ optical bistability (the NEPT). Light–matter coupling dominant ⇒ the lasing threshold (Route I). Both comparable ⇒ the coexistence region, the driven-dissipative counterpart of Zhang's superradiant solid.

This is a useful thing to be able to say in a paper: the proposal is not a competing claim about which transition is "the" right one, but a statement that the sensor should be operated in the light–matter-dominated corner of a phase diagram whose other corner is already known to work. It also sets the design rule — keep χ small (moderate n, or a Rydberg pair with a small van der Waals coefficient) unless the folded, latching response is wanted deliberately.

Neither Ref. [2] nor Ref. [28] addresses sensing; both are equilibrium phase-structure studies. Reference [30] is the closest existing work to Route I — a Rydberg medium in a cavity with a narrow transition, where the Rydberg nonlinearity "strongly modifies the superradiance threshold" — but its aim is generating ultranarrow nonclassical light, not measuring a field.

6 The three transitions compared

	NEPT (incumbent)	**Route I** ($U(1)$ lasing)	**Route II** ($\mathbb{Z}_2$ Dicke)
Scale factor	empirical, calibrated	$-d/2\hbar$ (exact)	$-d/2\hbar$ (exact)
Read-out	optical amplitude	optical *frequency*	amplitude / homodyne
Cavity-drift immunity	none	**rejected** $\kappa/\Gamma \approx 10^2$	none; drift amplified with signal
Critical point set by	density, interactions	repump, N, finesse	laser power (in situ, μs)
Reproducibility	drifts with T, density, stray fields	instrument setting	instrument setting
Hysteresis	in the bistable region, yes; none at the critical endpoint itself	no (second order)	no (second order)
Noise floor	empirical: 49, 2.6	$\sqrt{2\Gamma/N}$, all N atoms	$0.72\sqrt{\Gamma/n_b}$, $n_b \ll N$
At $N = 10^6$, matched Γ	—	0.070	0.358
Transduction gain	steep edge	threshold nonlinearity	ε^{-1}, 3 decades free
Squeezing	no	no (classical light)	-10.3 **dB, self-generated**
Bandwidth	fast (vapour)	$\sim\kappa$	$(\kappa+\gamma)/2 \approx 27.5$ kHz
Extra hardware	none	repump	second Raman channel
Maturity	**demonstrated twice**	apparatus exists [33]	engineering published [29]

Table 4: The three critical mechanisms at matched atomic decoherence Γ. Sensitivities in nV cm^{-1}Hz$^{-1/2}$.

Where Route I wins. Cavity-drift immunity, and it is not a small effect: the dominant technical noise channel is suppressed by κ/Γ rather than amplified. It also uses every atom, giving a floor a factor 5.1 below Route II at equal N, and it reads the measurand out as an optical frequency — the most precisely measurable quantity there is. It is also the cheaper experiment.

Where Route II wins. It is a genuine ground-state critical point, so the apparatus of critical quantum metrology applies to it; it delivers ε^{-1} transduction gain with a wide window free of bandwidth cost; and it generates its own squeezed light (-10.3 dB), which Route I — producing essentially classical laser light — does not. If the goal is a physics result rather than an instrument, Route II is the stronger claim.

Where both lose to the incumbent. The NEPT has been *built*, twice, in a hot vapour cell. Both routes here need a good optical cavity and, for Route I, a narrow transition plus repumping; Route II most likely needs cold atoms. Neither offers a better scaling law: all three go as $1/\sqrt{N}$, and Sec. 4 shows explicitly that criticality does not move that exponent. What the light–matter routes offer is a better *prefactor* and, more importantly, a critical point that is an instrument setting rather than a property of the sample.

Recommendation. Route I first: cheaper, lower floor, and its drift immunity is the single most defensible advantage over the incumbent. Route II is the follow-up, and where the interesting quantum-metrology physics lives.

7 Prior art

The relevant literature falls into four strands. The purpose of this section is to be honest about how much of the idea is already occupied.

7.1 Cavity-enhanced Rydberg electrometry — the direct lineage

Peng and co-workers have published a coherent programme along exactly this line. Reference [1], the seed paper, places a four-level Rydberg-EIT medium in a cavity and reports a minimum detectable field roughly 8× smaller than the cavity-free case and a spectral resolution improved by more than 20×, attributing both to cavity narrowing of the transmission features; it also notes robustness against control-field amplitude and broadband tunability. The same group then moved to the collective regime: Ref. [3] uses *collective Rabi splitting*, obtaining a four-peak cavity transmission whose outer splitting varies linearly with the microwave field, and reports about 7× improvement over plain EIT; Ref. [4] reports 196.7× over a bare atomic medium and 26.2× over weak coupling, a minimum detectable strength of 396.5 nV cm^{-1} (a detection floor, not a per-root-hertz sensitivity), and resolution gains of 216× and 10×. Experimentally, Ref. [5] demonstrates 18 dB of power-sensitivity enhancement using a microwave (not optical) resonator, and Ref. [33] demonstrates a cavity-enhanced Rydberg *superheterodyne* receiver with a 19 dB sensitivity gain.

Implication for the idea. The collective-coupling step is already taken. Anything proposed here must be distinguished from "$g_0\sqrt{N}$ in a cavity", because Ref. [3] is precisely that. The distinguishing feature has to be the *counter-rotating* terms and the phase transition they enable: on their published abstracts, Refs. [3, 4] are normal-phase Tavis–Cummings physics — a vacuum-Rabi/collective-Rabi splitting read out spectroscopically — with no transition and no critical point. Note this is exactly the same rotating-wave restriction found in Ref. [2] (Sec. 5), so the counter-rotating terms are the single ingredient that separates this proposal from the whole of that lineage.

7.2 Critical Rydberg metrology — the closest competitor

Ding, Liu, Shi, Guo, Mølmer and Adams [6] demonstrated many-body enhanced microwave metrology at the critical point of a non-equilibrium Rydberg gas, reporting a Fisher information three orders of magnitude above the independent-particle value and an equivalent sensitivity of 49 $\mathrm{nV\,cm^{-1}Hz^{-1/2}}$. Wang, Liang, Wang and co-workers [7] then put that critical system inside an optical cavity, reporting more than two orders of magnitude of additional Fisher information and 2.6 $\mathrm{nV\,cm^{-1}Hz^{-1/2}}$.

Both papers were read in full, and the mechanism is unambiguous. The critical point is that of a *non-equilibrium phase transition* (NEPT) — the optical bistability of a laser-driven Rydberg ensemble, arising from the competition between long-range Rydberg interactions, drive strength and dissipation. Feeding the interaction back as a population-dependent detuning, $\Delta_{\text{eff}} = \Delta - V\rho_{rr}$, makes the steady-state Bloch equations cubic in ρ_{rr}, and a cubic with three real roots is an S-curve. The two phases are the *non-interacting* one (low Rydberg density, $V\rho_{rr}$ negligible, atoms respond independently) and the *interacting* one (high density, collective response), with the Rydberg population as order parameter.

It is worth resolving an apparent inconsistency, because it is not sloppiness. The main text of Ref. [6] calls this "a second order dynamical phase transition", while its appendix derives bistability with a hysteresis loop — the signature of a *first*-order transition. Both are right, at different points of the same phase diagram: the structure is the liquid–gas one, a first-order line with a hysteretic bistable region terminating at a critical endpoint where the discontinuity vanishes and the susceptibility diverges. Ding *et al.* define the critical point as where $d\rho_{rr}/d\Delta \to \infty$ — the fold of the S-curve — and operate *at* that endpoint, where the transition is indeed continuous. Whether a driven-dissipative mean-field bistability is a phase transition in the thermodynamic sense is a separate and contested question; with finite-range interactions and atomic motion it may be replaced by fluctuation-driven switching between branches, with sharpness set by system size rather than a singularity. The sensor works either way, but the "phase transition" language is a soft spot. The "critical point" is operationally the coupling-laser frequency at which the transmission jumps, the jump direction depends on the scan direction, and a hysteresis loop is observed. Reference [7] works in a hot caesium vapour cell ($6S_{1/2} \to 6P_{3/2} \to 47D_{5/2}$, microwave to $48P_{3/2}$), uses the sharp bistability edge as a frequency discriminator, and quantifies it with the *classical* Fisher information; the cavity's role is to steepen that edge (by $> 10\times$ in slope, $> 100\times$ in Fisher information). No light–matter phase transition is involved anywhere in either work.

This is the strongest prior art and it must be addressed head-on. Reference [7] is "critical Rydberg metrology in a cavity", which is a fair one-line summary of the present idea too. The substantive difference is *which* criticality:

- Refs. [6, 7] use the *dissipative, Rydberg-interaction-driven* phase transition (optical bistability of the driven Rydberg ensemble). Its critical point is set by atomic density and interaction strength, and therefore drifts with temperature, density and stray fields, and cannot be tuned quickly. It is also hysteretic, so the discriminator is scan-direction dependent.
- The present proposal uses the *light–matter* (Dicke) transition. Its critical point is set by $\lambda \propto$ laser amplitude and by ω, both of which are tunable in situ on microsecond timescales and can be servoed. The critical point is a controlled instrument setting rather than a property of the sample.

The measurand couples differently too. Reading both papers turns up a second distinction that is arguably more important than the first, and that we had initially missed. *Neither uses Autler–Townes.* Ding *et al.* state that "the main effect of the microwave field here is to (1) induce a small AC Stark shift which moves the critical point, (2) change the population of the Rydberg atoms"; Wang *et al.* drive $47D_{5/2} \to 48P_{3/2}$ detuned by $\Delta_{\text{MW}}/2\pi = -200$ MHz, where AT splitting is simply unavailable, and describe the resulting "a.c. Stark shift. . . reflected on the NEPT signal as a small frequency shift". Three consequences follow:

- The response is *quadratic*, $\delta \approx \Omega_\mu^2/4\Delta_{\mathrm{MW}} \propto E_\mu^2$, not linear.
- A quadratic response has zero slope at zero field, so both experiments must bias at a nonzero amplitude — ≈ 2 mV/cm in Ref. [6], 4.87 (free space) and 4.95 mV/cm (cavity) in Ref. [7], in both cases tunable via the coupling detuning. Their quoted sensitivities are local slopes about that bias.
- The scale factor involves the Rydberg polarizability, the microwave detuning, the interaction strength and the position of the critical point. It is fitted ($k_c = 0.719$, $k_f = 0.035$), and neither paper claims SI traceability.

Both routes proposed here instead keep the *resonant AT dressing*, so the measurand enters linearly through $\tilde{\omega}_0 = \delta + \Omega_\mu/2$ with the exact, dipole-referenced scale factor $-d/2\hbar$. *The incumbent gives up AT traceability in order to get criticality; these routes keep both.* One caveat: resolving one dressed state requires $\Omega_\mu \gtrsim \Gamma$, so for very weak fields one falls back to Protocol C, where the microwave local oscillator supplies the bias — structurally the same move Refs. [6, 7] make, but with a linear rather than quadratic response about it.

Both differences are real and worth a paper, but they are differences of *controllability and traceability*, not of sensitivity scaling. Claiming the latter would be wrong (see Sec. 4.3.3).

7.3 Critical quantum metrology — the theory that constrains the claims

Garbe *et al.* [17] established the finite-component critical metrology paradigm with the quantum Rabi model, and Ref. [18] worked out the fully-connected-model scaling (static, quench and finite-time-ramp protocols, including Kibble–Zurek scaling) whose exponents we reproduce in Fig. 5. Against these, Rams *et al.* [19], Gietka *et al.* [20, 21] and Gyhm *et al.* [22] constrain what can be claimed once time is counted as a resource. Any write-up of this idea must engage with that second group of references explicitly; a referee will otherwise do it for you.

7.4 Dicke-model engineering and Rydberg cavity QED — the feasibility base

The engineered Dicke model is experimentally established: proposed by Dimer *et al.* [11], realised by Baden *et al.* [12] with cavity-assisted Raman transitions, and separately by Baumann *et al.* [13] via self-organisation of a BEC. For the Rydberg platform specifically, Ref. [29] gives a periodic-driving recipe for the anisotropic Dicke model in a cavity-coupled Rydberg array with a counter-rotating/rotating ratio tunable from zero to infinity, and Ref. [28] maps the resulting phase diagram by quantum Monte Carlo. The enabling theory is therefore in place; what is missing is the sensing application. Cavity Rydberg polaritons have been observed [23], including the suppression of inhomogeneous broadening by collective strong coupling that this proposal needs. On the sensing side, Rydberg electrometers now approach the standard quantum limit in cold atoms [24] and reach within 1.0 dB of it in tweezer arrays [25], and collective-radiative linewidth engineering for Rydberg microwave sensing has just been proposed [26]. Rydberg systems can also be spin-squeezed directly [27], which is the main competing route to sub-SQL electrometry.

7.5 What appears to be unoccupied

An arXiv full-text search combining *Dicke / superradiant / lasing threshold / cavity pulling / bad cavity* with *Rydberg*, *electrometry*, *microwave electric field*, *metrology* and *sensing* returns no hit that uses a light–matter transition as the sensing element for microwave electrometry. The Dicke–Rydberg literature (Refs. [2, 29, 28, 37]) is uniformly concerned with phase structure and dynamics; the critical-electrometry literature (Refs. [6, 7]) is uniformly concerned with the interaction-driven NEPT; and Ref. [30], which does combine a Rydberg medium with a superradiance threshold in a cavity, aims at generating nonclassical light. On that basis:

1. **Unoccupied.** Using a *light–matter* transition — either flavour — as the critical resource for microwave electrometry, in place of the Rydberg-interaction transition of Refs. [6, 7].
2. **Unoccupied, and the strongest single result.** The cavity-pulling read-out of Sec. 3: reading the microwave field as the frequency of a bad-cavity laser, so that cavity drift is rejected by κ/Γ instead of amplified. The underlying mechanism is standard active-optical-clock physics; what is new is applying it to electrometry, where drift amplification is the known weakness of every critical scheme.
3. **Unoccupied, but with near neighbours.** The null / threshold-crossing read-out. Biasing at a dynamical threshold and reading a frequency is not new in itself: Ref. [35] pulls a self-sustained microwave oscillator through its Adler synchronisation boundary, and Ref. [36] exploits a sub- to superradiant emission threshold for atomic state read-out. The novelty claim should be worded at the level of the specific realisation, not as "threshold-based sensing".
4. **Unoccupied, and useful.** The observation of Sec. 5.1 that the incumbent NEPT is recovered from the lasing route by turning up the Rydberg interaction — i.e. that the two critical mechanisms are corners of a single phase diagram — and the gain–bandwidth characterisation of Sec. 4.

Item 2 is the one to lead with.

8 Assessment

8.1 Is it a good idea?

Yes, provided it is framed as a substitution rather than a scaling claim. The physics is sound, the map from measurand to phase boundary is exact and simple in both routes, the ingredients exist separately, and the numbers are competitive with — for Route I, better than — the incumbent.

What does *not* survive scrutiny is the framing "criticality gives super-classical scaling for electrometry". Equations (24) and (12) show, for the two routes independently, that the floor is the ordinary $1/\sqrt{N}$ standard quantum limit; Sec. 4 shows that the $N^{4/3}$ ground-state advantage is removed by critical slowing down, in agreement with the known no-gos.

What does survive is sharper and more useful. Against the incumbent NEPT the light–matter routes offer (i) a critical point that is an *instrument setting* — repump and finesse, or laser power — rather than a property of the sample that drifts with temperature and density; (ii) an exact rather than calibrated scale factor; and, for Route I, (iii) active rejection of the dominant technical noise channel by $\kappa/\Gamma \approx 10^2$, which is the first answer anyone in this field has to the standing objection that criticality amplifies drift along with signal. Framed that way it is a good idea, and Route I is the one to build.

8.2 Benefits

- **Cavity-drift rejection (Route I).** $\kappa/\Gamma \approx 10^2$ suppression of the dominant technical channel, by the bad-cavity mechanism that underpins active optical clocks. This is the strongest single argument for the substitution.
- **A frequency read-out (Route I).** The measurand appears as an optical frequency at 1.124 MHz per (mV/cm), not as an amplitude — and all N atoms participate, giving a floor 5.1× below Route II at equal N.
- **Linear, dipole-traceable coupling (both routes).** The measurand enters through resonant AT dressing, so the scale factor is exactly $-d/2\hbar$. The incumbent NEPT couples the field through a *quadratic* AC Stark shift about a nonzero bias, with a fitted scale factor and no traceability claim (Sec. 7).

- **Transduction gain (Route II).** ε^{-1} gain, with roughly three decades of it free of bandwidth cost. Real Rydberg sensors are usually limited by optical detection and technical noise, not by atomic projection noise, so gain at the atomic stage is directly valuable.
- **In-situ tunable critical point.** Unlike the density-driven critical point of Refs. [6, 7], the boundary here is set by repump and finesse (Route I) or laser power (Route II) and can be servoed, making the operating point stable and reproducible.
- **SI traceability via Protocol B.** Reading the measurand as a laser detuning preserves the metrological virtue that makes Rydberg electrometry attractive to national metrology institutes.
- **Self-generated squeezing (Route II).** Up to 10 dB, with no external squeezed source and no separate entanglement-generation step. Route I, producing classical laser light, does not offer this.
- **One phase diagram, two corners.** The incumbent NEPT is recovered from Route I by turning up the Rydberg interaction (Sec. 5.1), so the proposal is a move within a known phase diagram rather than a competing claim.
- **Bright-to-dark operation.** The response sign is favourable for detection.
- **A trigger mode.** With strong Rydberg interactions the transition can be made first order, giving a latching threshold detector.

8.3 Risks and objections

- **The scaling advantage is illusory.** Established in Sec. 4.3.3 and by Refs. [19, 20, 22]. This is the objection a referee will raise first.
- **Route I's inversion is unproven (Route I only).** The route assumes lasing on a ground-to-dressed-Rydberg transition; no superradiant laser has been built on anything but a narrow optical transition. See Sec. 3 for the alternative architecture and what it would cost. This is the single largest open question for Route I, and it is the reason Route II may be the safer target despite being the more demanding experiment.
- **Criticality amplifies drift as well as signal.** Protocol A has no discrimination between $\delta\tilde{\omega}_0$ from the microwave field and $\delta\lambda$ from laser intensity noise or $\delta\omega$ from cavity drift — formally, the critical QFI matrix is sloppy [34]. Protocol C's IF encoding and Protocol B's null lock are the mitigations, and they are not optional.
- **Bandwidth.** 27.5 kHz at the parameters used, set by $(\kappa+\gamma)/2$. That is fine for metrology and useless for a communications receiver. A higher-κ cavity trades gain for bandwidth in the usual way.
- **Rydberg blockade caps N_{eff}**, which is what Eq. (24) actually depends on. This is in direct tension with the Ref. [2] physics, which *wants* strong interactions.
- **Engineering overhead.** Two Raman channels plus a high-finesse cavity plus (probably) cold atoms is a substantial step away from the vapour-cell simplicity that makes Rydberg electrometry commercially interesting — though less of a step than it was, given Refs. [29, 33].
- **Decoherence cuts off the critical scaling.** The divergence is truncated once the soft-mode frequency falls below γ, i.e. at $\varepsilon_{\min} \sim (\gamma/\omega)^2$; with vapour-cell linewidths this bites early.
- **Prior art is close.** Reference [7] in particular will be cited against any claim of novelty that is not carefully worded.

8.4 Recommended programme

1. **Settle Route I's inversion question first.** Decide between lasing directly on the dressed-Rydberg transition and lasing on a narrow optical transition that the Rydberg admixture shifts (Sec. 3). The sensitivity numbers depend on the answer, and everything else about Route I is downstream of it. If the answer is unfavourable, go to Route II, which needs no inversion.

2. **Then build Route I.** It is the cheaper experiment, it reaches the lower floor, and its drift immunity is the defensible advantage. The apparatus already exists: Ref. [33] is a cavity-enhanced Rydberg superheterodyne receiver with a 19 dB gain, and what it lacks is a repump and a narrow transition. The first measurement to make is simply Eq. (6) — confirm that the emitted frequency tracks $\tilde{\omega}_0$ and ignores the cavity.

3. **Theory paper, correctly framed.** Something like "Microwave electrometry at a light–matter phase transition". Lead with the substitution (Table 1), the drift-rejection result and the bridge back to the NEPT (Sec. 5.1); state the no-go results on critical scaling up front rather than being caught by them.

4. **Do the calculations a referee will ask for.** A Ziv–Zakai or Bayesian treatment of threshold-location precision, since the Cramér–Rao bound is the wrong tool for a null measurement; a proper account of inhomogeneous coupling across the cavity mode; and a full quantum (rather than semiclassical Maxwell–Bloch) treatment of Route I's linewidth, since the mHz numbers in Sec. 3 rest on the standard bad-cavity Schawlow–Townes expression rather than on a calculation done here.

5. **Address the selectivity objection formally.** That criticality amplifies $\delta\lambda$ and $\delta\omega$ as readily as $\delta\tilde{\omega}_0$ is, in estimation language, the statement that the critical Fisher-information matrix is sloppy. Reference [34] shows for the Dicke model that multi-parameter critical estimation can retain divergent precision at the cost of a reduced exponent, and that a Dicke dimer with photon hopping restores the optimal scaling. Route I answers a large part of this objection physically rather than statistically, which is worth saying.

6. **Then Route II**, following the anisotropic-Dicke recipe of Ref. [29], for the ground-state critical point, the ε^{-1} gain and the squeezing.

A Numerical methods

All code is in `code/`; all figures are regenerated by `code/make_figs.py`.

Exact diagonalisation. Equation (13) is built on the product of the $j = N/2$ spin manifold ($N+1$ states) and a truncated Fock space. Since $\partial H/\partial\tilde{\omega}_0 = J_z$ is parity-even, only the even-parity sector contributes to F_Q and the calculation is restricted to it, which also removes the closing odd-parity gap from the eigensolver's path. Photon truncation was verified at $N = 200$, $\lambda = \lambda_c$: F_Q agrees to twelve digits between $n_{\max} = 60$ and $n_{\max} = 120$; production runs used $n_{\max} = 100$–140. The QFI is obtained from the fidelity susceptibility with a symmetric step $h = 10^{-4}\tilde{\omega}_0$.

Quantum Langevin. The linearised open-system calculation uses symmetrised two-sided spectra with vacuum inputs, diffusion matrix $D = \mathrm{diag}(\kappa, \kappa, \gamma, \gamma)$, and the input–output relation $x_{\text{out}} = \sqrt{2\kappa}x_a - x_{\text{in}}$ including the cross-correlation between intracavity field and input vacuum. The implementation returns exactly $S_{\text{out}} = 1/2$ for an empty cavity, which fixes all normalisation conventions. The damped threshold λ_c is located by bisection on $\max\mathrm{Re}\,\mathrm{eig}(A) = 0$. Homodyne angles are optimised on a 361-point grid.

Parameters. Unless stated: $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$ MHz, $\kappa/2\pi = 50$ kHz, $\gamma/2\pi = 5$ kHz, $d = 1774.82\, ea_0$ for ^{87}Rb $53D_{5/2} \to 54P_{3/2}$.

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